Machine Learning – Supervised Regression

Course Outline

Session 1 Basics of Machine Learning & Data Pre-processing

Basics of Machine Learning

- Machine Learning Overview
- Traditional Programming Vs Machine Learning
- Understanding the Problem and Data
- Steps in Machine Learning
- Basic Terms used in Machine Learning
- Types of Machine Learning

Data Preprocessing

- Missing Values (Standard Missing Values, Non-Standard Missing Values)
- Handle Non-Numeric Data (One-Hot Encoding, Label Encoding, Ordinal Encoding)
- Normalization and Transformation
- Outliers Detection/removal (Based on Boxplot, IQR, Z-score, Scatter plot)
- Introduction to Feature Engineering
- Train Test Split

Session 2 Basics of Machine Learning & Linear Regression

- Applications of Machine Learning : Use Cases
- Linear Regression
  - Simple linear Regression
  - Multiple Linear Regression
  - Visiting basics: Covariance & Correlation
  - Regression Analysis
  - Ordinary Least square Method
  - Measure of Variation : R-squared & Adjusted R-Squared
  - Inferences about slope

- Linear Regression with time series data :Autoregression

Session 3 Assumptions of Linear Regression and Model Evaluation

- Assumptions of Linear Regression
- Tests for assumptions of Linear Regression
- Model evaluation Metrics
- Presence of categorical variable
- Interaction effect

Session 4 Feature Engineering

- Feature Extraction
- Feature Transformation
- Feature Scaling
- Feature Selection
    - Forward Selection
    - Backward elimination
    - Stepwise selection/regression
    - Recursive Feature Elimination (RFE)

Session 5 Model Optimization

- Bias and Variance
- Overfitting and Underfitting
- Model Validation
- Gradient Descent
- Regularization
    - Lasso
    - Ridge

  Hyperparameter Tuning
    - Grid Search
    - Random Search
    - Bayesian Optimization

Session 6 Deployment & Case Study

- Serializing machine learning models
- Exposing the model through Rest APIs
- Packaging for reproducibility
- Create ML pipeline
- Scaling the model